

Module 2 : Levels of NLP Processing

Morphological Analysis

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Agenda

- Revision : NLP Terms
- Morphological Analysis -
- English Morphology,
- Derivational and
- Inflectional Morphology
- Finite State Transition (FST)

Revision of Module 1

- What is NLP and Why it is Hard?
- Mention few applications of NLP in various Sectors.
- Describe the stages of NLP
- What are the main challenges of NLP in all levels ?
- What are the major components of NLP and explain?
- Differentiate between Natural language and Computer language.
- Explain the different levels of NLP.
- Write down the difference between NLU and NLG with a suitable example

NLP Revision (Cont'd)

Given an image, can our machine answers the questions in natural language?



What color are her eyes?
What is the mustache made of?



How many slices of pizza are there?
Is this a vegetarian pizza?



Is this person expecting company?
What is just under the tree?



Does it appear to be rainy?
Does this person have 20/20 vision?

NLP Revision (Cont'd)



Natural
Language
Understanding

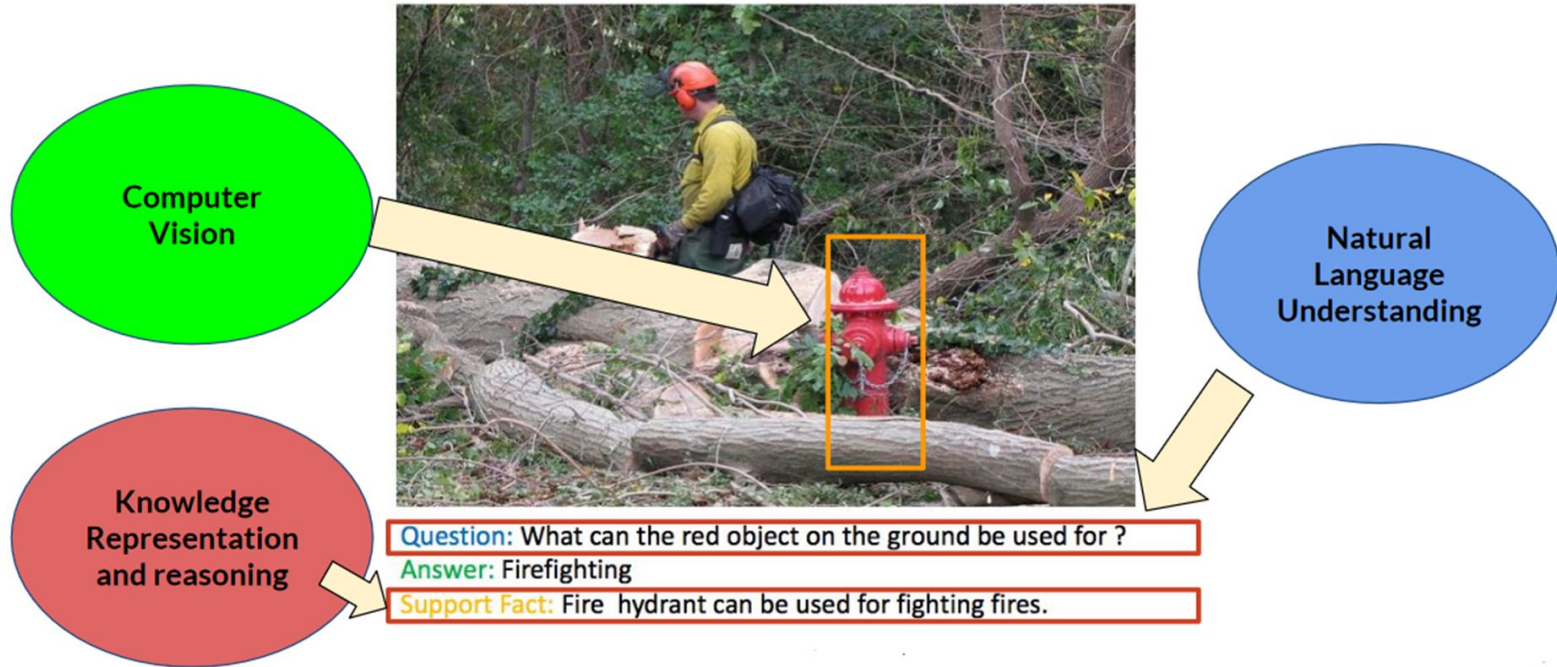
Question: What can the red object on the ground be used for ?

Answer: Firefighting

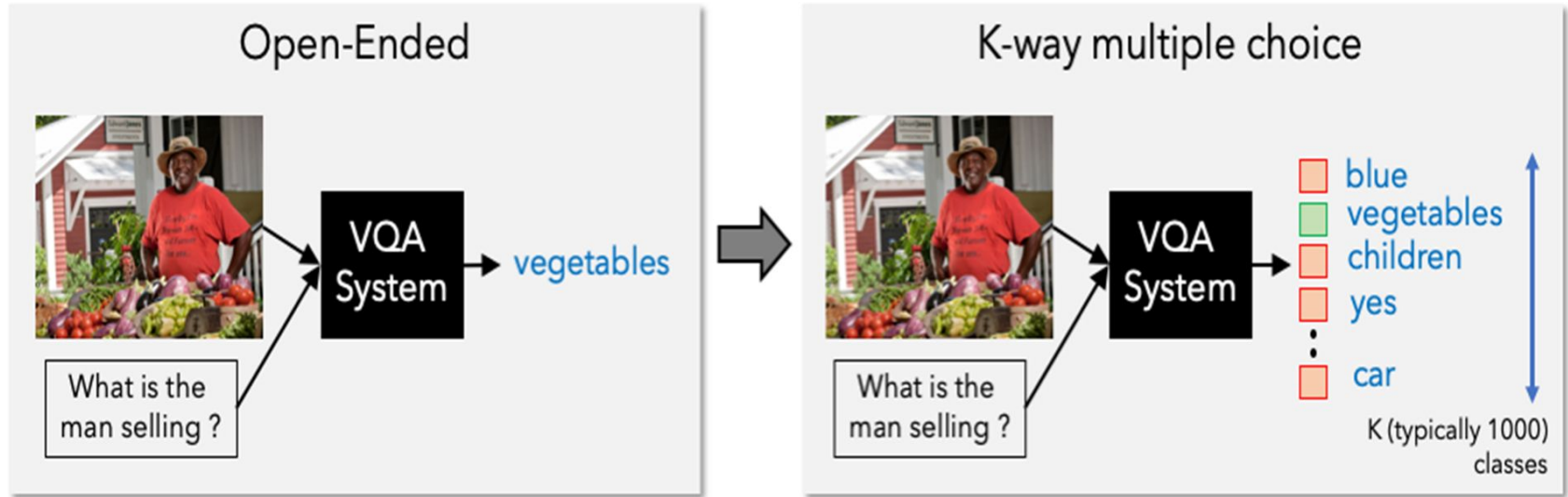
NLP Revision (Cont'd)



NLP Revision (Cont'd)



NLP: Visual Question Answering (VQA)



VQA System Example : MS CoCO QA

- ❑ Automatically generate QA pairs with MS COCO captions
- ❑ 123,287 images (72,783 for training and 38,948 for testing) and each image has one QA pair.
- ❑ 4 types of question templates: What object, How many, What color, Where



COCOQA 5078
How many leftover donuts is the red bicycle holding?
Ground truth: three

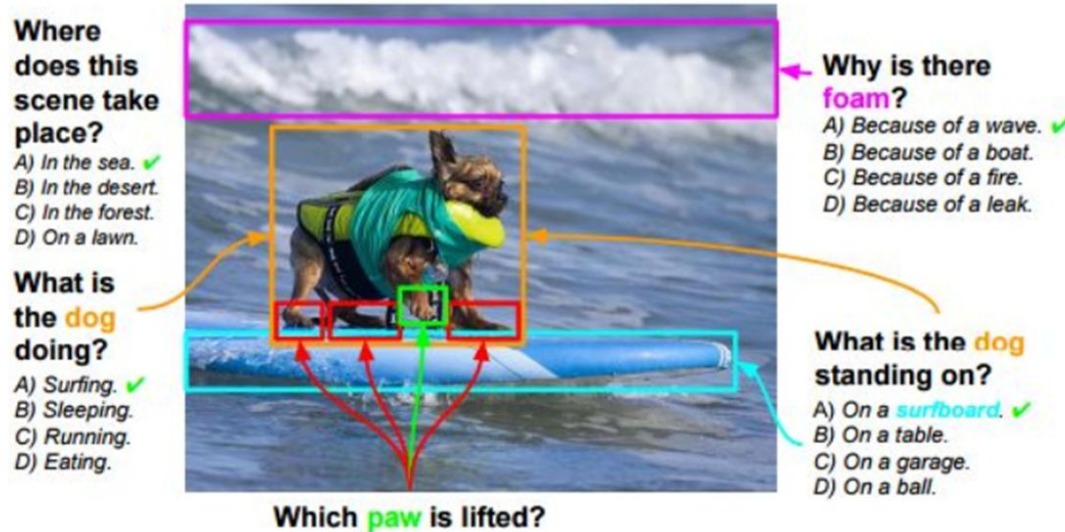


COCOQA 1238
What is the color of the tee-shirt?
Ground truth: blue

VQA System Example : 7W VQA

- 47,300 COCO images, 327,939 QA pairs, and 1,311,756 human-generated multiple-choices
- 7W stands for what, where, when, who, why, how and which

Telling
Pointing



VQA System Example: [Github Link](#)

Vehicles and Transportation



Q: What sort of vehicle uses this item?

A: firetruck

Brands, Companies and Products



Q: When was the soft drink company shown first created?

A: 1898

Objects, Material and Clothing



Q: What is the material used to make the vessels in this picture?

A: copper

Sports and Recreation



Q: What is the sports position of the man in the orange shirt?

A: goalie

Cooking and Food



Q: What is the name of the object used to eat this food?

A: chopsticks

Geography, History, Language and Culture



Q: What days might I most commonly go to this building?

A: Sunday

People and Everyday Life



Q: Is this photo from the 50's or the 90's?

A: 50's

Plants and Animals



Q: What phylum does this animal belong to?

A: chordate, chordata

Science and Technology



Q: How many chromosomes do these creatures have?

A: 23

Weather and Climate



Q: What is the warmest outdoor temperature at which this kind of weather can happen?

A: 32 degrees

Features of Existing VQA System

Dataset	Number of QA pairs	Annotation	Question Diversity	Answer Diversity (top-1000 answers coverage)
COCO-QA	117,684	Open-ended	Low (generated from captions)	100%
VQA	614,163	Open-ended +Multiple choices	Encourage to have diverse annotations	82.7%
Visual 7W	327,939	Multiple choices	At least 3 W for each image	63.5%

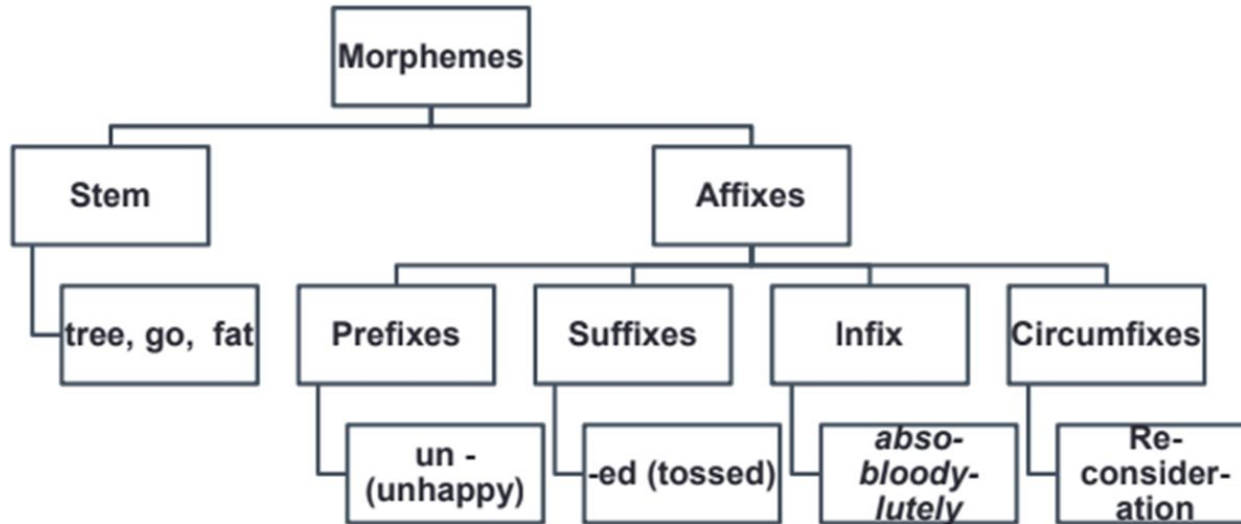
Morphology: Introduction

- Morphology is the scientific **study of word formation**.
- **Minimal meaningful / grammatical unit** of a word.
- The **study of patterns of word** formation.
- How words are originated, **grammatical forms**, on what basis the word class is formed etc.
- All Languages make new words from existing resources / words.
- Eg. : Beautiful, Ladies Tailor, Walking, Sitting etc.

Morphology: Introduction

- Word formation **rules from root words**
- First **crucial step** in NLP
- Languages **rich in morphology**: e.g., Dravidian, Hungarian, Turkish
- Languages **poor in morphology**: Chinese, English
- A task of interest to computer science: **Finite State**

Morphology: Introduction



Morphology: Introduction

Morphology studies the internal structure of words, how words are built up from smaller meaningful units called **morphemes**

- Their **internal structure**

washing → wash + -ing

- How they are **formed**?

bat	→	bats	∴	rat	→	rats
write	→	writer	∴	browse	→	browser

- **Morphology tries to formulate rules**

What is Morph then?

- A **morph is the phonetic realization** of a morpheme.
- The real form of the morpheme, the **actual utterance of the morpheme.**
- A **formal unit** with a physical shape.
- This is **similar to** the concept called '**phone**' in phonology



Allomorphs

- **Variants** of the same morpheme, but **cannot be replaced** by one another
- **Example**
- **opposite:** un-happy, in-comprehensible, im-possible, ir-rational

Allomorphs

- Allomorph is **phonologically distinct variants** of the same morpheme.
- **Different realization, manifestation** of the same morpheme.
- They **vary in shape or pronunciation** according to their condition of use.
- They are a **class of morphs** which are semantically identical.
- They are **derived from phonological and morphophonemic rules**.

What is Word?

- “**Word**” is a popularly acknowledged notion
- A **meaningful unit of speech** which is normally uninterruptible and which when written or printed has spaces on either side
- A sound or group of **sounds that expresses a meaning and forms** an independent unit of language
- A minimal **free form, or a “*morpheme*”**

Technological developments bring in new terms, additional meanings/nuances for existing terms

- **Justify** as in *justify the right margin* (word processing context)
- **Xeroxed**: a new verb
- **Digital Trace**: a new expression
- **Communifaking**: pretending to talk on mobile when you are actually not
- **Helicopter Parenting**: over parenting



Components of Morphology

- **Free Morpheme** : It can occur by itself and has meaning. It can be either Lexical free or Functional free Morphemes.
- Eg. : Home, November, to, the, etc.
- **Bound Morpheme** : un, al, -s, -ed, -er, -est
- **Root / Stem + Affix** : home + ly, Per + mit, good+ est = Best



Lexical Free Morphemes

(Carry real-world meaning)

Word

Free Morpheme

book

book

run

run

happy

happy

child

child

computer

computer

write

write

fast

fast

beauty

beauty

food

food

play

play

Functional Free Morphemes

(Grammatical meaning, limited vocabulary)

Type	Examples
Articles	a, an, the
Pronouns	he, she, they, it
Prepositions	in, on, at, under
Conjunctions	and, but, or
Determiners	this, that, these
Auxiliary verbs	is, are, was, have

Bound and Free Morphemes

- **Bound**

Cannot appear as a word by itself.

-s (dog-s), -ly (quick-ly), -ed (walk-ed)

- **Free**

Can appear as a word by itself; often can combine with other morphemes too.

house (house-s), walk (walk-ed), of, the, or

Classification of Bound Morphemes

Bound morphemes are classified into:

1. **Derivational Morphemes**
2. **Inflectional Morphemes**

This **classification is based on:**

- Change in meaning
- Change in word class
- Grammatical function

Derivational Morphology

- **Derivational morphology** deals with the formation of **new words**
- Derivational morphemes change the **meaning or grammatical category** of a word

Example:

- happy (Adj) → happiness (Noun)
- write (Verb) → writer (Noun)

Derivational Morphemes

- Create new words
 - May change word class
 - Change the meaning of the base word
 - Can be prefixes or suffixes
 - Occur before inflectional morphemes
- Derivational prefixes modify meaning but usually do not change word class

Examples:

- un- → unhappy
- re- → rewrite
- dis- → disagree
- pre- → preview
- mis- → misunderstand

Derivational Morphemes

- Derivational suffixes **often change the word class**

Examples:

- -ness: happy → happiness (Adj → Noun)
- -er: teach → teacher (Verb → Noun)
- -able: read → readable (Verb → Adj)
- -ment: develop → development (Verb → Noun)

Derivational Morphemes

- **Word Formation Using Derivation**

Examples:

- nation → national → international
- beauty → beautiful → beautifully
- govern → government

Each step forms a **new lexical item**

Inflectional Morphology

- **Inflectional morphology** deals with **grammatical variations** of words
- Inflectional morphemes do **not create new words**
- They provide grammatical information

Example:

- walk → walked
- cat → cats

Inflectional Morphology

- Do not change word class
- Do not change core meaning
- Indicate grammatical features
- Always occur **after derivational morphemes**
- Limited in number (in English)

English has **8 inflectional morphemes**:

- **Noun**: -s (plural), -'s (possessive)
- **Verb**: -s, -ed, -ing, -en
- **Adjective**: -er, -est

Order of Morphemes

Rule: **Root** → **Derivational** → **Inflectional**

Examples:

- **teach + er + s** → **teachers**
- **happy + ness + es** → **happinesses**

Incorrect:

- **teach + s + er** ❌

Inflectional Morphology: Examples

Word Class	Inflectional Morpheme	Example Word	Function
Noun	-s / -es	books, classes	plural
Noun	-’s	teacher’s	possession
Verb	-s	runs	third person singular
Verb	-ed	walked	past tense
Verb	-ing	running	present participle
Verb	-en	written, taken	past participle
Adjective	-er	taller	comparative degree
Adjective	-est	tallest	superlative degree

Inflectional and Derivational Morphology

- Two different kind of relationship among words
- ***Inflectional morphology***
 - Grammatical:
number, tense, case, gender
 - Creates new forms of the same word:
bring, brought, brings, bringing
- ***Derivational morphology***
 - Creates new words by changing part-of-speech:
logic, logical, illogical, illogicality, logician
 - Fairly systematic but some derivations missing:
sincere - sincerity, scarce - scarcity, curious - curiosity, fierce - fierceness?

Inflection Morphology : Summary

- **Modifies a word** in order to express **different grammatical categories** such as **tense, mood, voice, aspect, person, number, gender, case**
- **Verbal inflection:** conjugation - **walks, walked, walking**
- **Nominal inflection:** declension - **computers**
- Usually realised by
 - prefixation (Prefix)
 - suffixation (Suffix)
 - circumfixation **ge+hab+t**
 - reduplication: **orang+orang** (plural of „man“ in Indonesian)

Derivation Morphology : Summary

- Creates new words
- **Examples:** un+translat+abil+ity, piti+less-ness
- Changes the part-of-speech and/or meaning of the word
- Adds prefixes, suffixes, circumfixes
- **Conversion:** changes the part-of-speech without modifying the word:
 - book (N) → book (V), leid(en) (V) → Leid (N)

Root, Base and Stem

- **Root:** A root is the irreducible core of a word, with absolutely nothing else attached to it. *E.g. jump- jumps, jumping, jumped.* Here, jump is the root.
- **Stem:** the stem is that part of a word that exists before the addition of any inflectional morpheme. *E.g. worker -> workers, shift -> shifted*
- **Base:** Base is any unit of a word where any kind of affixes can be added. It could be both inflectional or derivational. *E.g. boy -> boys, boy -> boyish, boy -> boyhood*

Root, Base and Stem

- **Root:** The smallest meaningful part of a word.
- **Base:** Any unit to which affixes can be attached.
- **Stem:** The form of a word before inflectional endings.

Example 1: unhappiness

- **Root:** happy
- **Base:** unhappy
- **Stem:** unhappy

Explanation:

happy is the core meaning unit (root).

unhappy is the form to which -ness is added (base).

Since there is no inflection, the stem is also unhappy.

Root, Base and Stem

Example 2: replaying

- **Root:** play
- **Base:** replay
- **Stem:** replay

Root → core meaning

Base → form for adding affixes

Stem → form before inflection

Explanation:

play is the root.

replay is the base for adding -ing.

replay is also the stem because -ing is an inflectional suffix.

Types of Affixes

- **Prefix:** un-, anti-, etc (a-, ati-, pra- etc.)
 - *un-happy, pre-existing*
- **Suffix:** -ity, -ation, etc (-taa, -ke, -ka etc.)
 - *talk-ing, quick-ly*
- **Infix:**
 - *'n' in 'vindati' (he knows), as contrasted with vid (to know)*
 - **English: 'Fan-Flaming-Tastic.'** (emphasis)
 - E.g. (Roya Nikkhah, "Prince William's Nanny [Tiggy Pettifer] Says Engagement Is 'Fan-Flaming-Tastic.'" The Telegraph [UK], Nov. 21, 2010)*
- **Circumfixes** - precedes and follows the stem
 - **English: re-consider-ation**

Ingredients of a Morphological analyser

- List of roots with Part-of-Speech.
- List of derivational affixes
- Morphotactics Rules
- Orthographic (Phonological) Rules

Morphotactics in Action

- The rules governing the ordering and combination of morphemes (stems and affixes) to form valid words in a language.
- **Purpose:** It dictates which class of morpheme can follow another (e.g., a plural marker must follow a noun stem, not a verb stem)
- **Role in NLP:** It is a core component of morphological analyzers and generators, often implemented using Finite-State Automata (FSA).
- **Focus:** It answers "Does this combination of parts make sense?"
- Eg.: Which morphemes can be arranged in which order?
 - cat, walk, translat+able

Morphotactics in Action: Example

- **English Nominal Inflection:** [Noun Stem] + [Plural Suffix -s/-es]
 - Valid: dog + s -> dogs
 - Invalid : s + dog -> (Affixes cannot precede stems in English)
- **English Verb Inflection:** [Verb Stem] + [Tense/Aspect]
 - Valid: walk + ed -> walked
 - Valid : walk + ing -> walking
- **Complex Example:** English affixes may only be attached directly to morphemes with particular parts of speech:
 - *do + -able + -ity = doability, translat + abil + ity, translat + able*

but not : *do + -ity + -able = *doityable, translat + ity + abil, translat + able + ity*
(allomorphs *able - abil*)

The suffix *-ity* produces a noun from an adjective, and *-able* creates adjectives from verbs.

Morphotactics in Action: Example

- **Hindi Noun Inflection: Root + Gender/Number Suffix + Case Marker (Postposition).**
- **"लड़कों" (Ladkon - Boys, Oblique Plural)**
 - **Root:** लड़क (Ldak)
 - **Suffix:** -ों (-on, plural oblique marker)
 - *Note:* The masculine singular "लड़का" (Ladka) changes to "लड़के" (Ladke - plural direct) or "लड़कों" (oblique).
- **Verbal Morphotactics (Verb Inflection) :** Hindi verbs combine a root with aspect and tense markers.
- **Verb Root+Aspect/TAM+Gender/Number/Person**
- **Example: "जाएगा" (Jaayega - Will go, Masculine, Singular)**
 - **Root:** जा (Ja)
 - **Suffix:** -एगा (-ega, future tense masculine)

Morphotactics in Action: Example

- **Derivational Morphotactics (Word Formation)** : Adding suffixes to create new word classes (e.g., Verb to Noun)
- **Example:** "लिखाई" (Likhaee - Writing/handwriting)
 - **Root:** लिख (Likh - verb)
 - **Suffix:** -आई (-aaee - turns verb into noun)
- **Case Marker (Postposition) Rules** : Hindi uses postpositions (like "ne," "ko," "se") rather than prepositions, which are placed *after* the modified root
- **Rule:** If a noun is followed by a postposition, it must take the oblique form.
- **Example:** लड़का (Boy) + ने (ergative) = लड़के ने (Ladke ne - By the boy)
- **Example:** घर (House) + में (in) = घर में (Ghar mein - In the house)
- **Compounding (Compound Verbs)** : Hindi often combines two verbs to create a new, refined meaning.

Morphotactics in Action: Example

- **Compounding (Compound Verbs) : Example**
- **Example: "खिलाना" (Khilaana - To feed/To make eat)**
 - **Root:** खा (Kha - eat)
 - **Suffix:** -आना (-aana - causative)
- **Example: "पढ़कर" (Padhkar - Having read)**
 - **Root:** पढ़ (Padh)
 - **Suffix:** -कर (-kar - conjunctive participle)

Orthographic Rules

- Define additional FSTs to implement spelling rules:
 - **Gemination**: consonant doubling (beg -> begging),
 - **Elision**: 'e' deletion (make -> making),
 - **Epenthesis**: 'e' insertion (fox -> foxes), etc.
 - **Y replacement**: 'y' changes to 'ie' before 'ed' (try -> tries)
 - **I spelling**: 'I' changes to 'y' before a vowel (lie -> lying)
- Spelling rules: $e \rightarrow e / x \wedge _ s$ (eg. **fox^s -> foxes**)
- Multiple spelling rules might apply for a given word.
- **Eg**: **spy^s -> spies** (y->i, e->e)
- All spelling rules execute in parallel.
- Rewrite rules can be compiled into FSTs.